Abu Dhabi City Campus Daily/Weekly Planner

مدارس الإمارات الوطنية
Emirates National Schools

Week 12: 16–20 November 2026

Subject: Programming **Unit Title: Python Calculations & Debugging** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1 Python arithmetic operators	Learning Target: I can use +, -, * and / to calculate and display results in Python. Focus: review numeric variables, print() and simple arithmetic.	Formative: predict output and quick exit ticket. Support/LLD: match operator symbols to words and code icons; no arithmetic. Core: run four short calculations. Extension: explain changed outputs.	Python editor/Programiz; operator symbol cards; worked examples; guided worksheet.
Lesson 2 Input and calculations	Learning Target: I can use input(), int() or float() and arithmetic operators in a simple interactive program. Focus: why numeric input must be converted.	Formative: code-along, two test inputs and peer check. Support/LLD: order illustrated input/print blocks; MCQ. Core: build a total calculator. Extension: use float() with prices.	Python editor; data-conversion cheat sheet; calculator starter code; test checklist.
Lesson 3 Debugging calculation programs	Learning Target: I can locate, correct and test common errors in numeric Python programs. Focus: missing parentheses, wrong variable names, input conversion and incorrect operators.	Formative: find-the-bug pairs and exit ticket. Support/LLD: select correct code picture; match error icons. Core: fix three short programs. Extension: explain errors and add tests.	Fictional broken-code examples; debugging checklist; Python editor; teacher solutions.
Lesson 4 Create a UAE mini-calculator	Learning Target: I can independently create and test a small Python program that solves an everyday problem. Focus: variables, input, conversion, arithmetic and clear output.	Formative: program demo, partner testing and reflection. Support/LLD: visual program sequence and MCQ. Core: create UAE shopping or travel calculator. Extension: add if/else budget message.	Python editor; UAE-themed fictional shopping scenarios; simple success checklist; LMS submission.
Homework	Complete or improve a short calculator with named variables, input(), numeric conversion and print(). Show two successful test runs.	Formative: code review, test evidence and brief reflection.	Python editor; LMS task instructions; arithmetic-operator reference sheet.
Teacher's notes / Reminders	PROVISIONAL teacher-designed progression: an official Grade 12 Programming Week 12 scope and sequence was not supplied. Topics follow recently taught variables, input, data conversion and basic calculations. Confirm standards and formal assessments before submission.		

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Technology Integration	Programiz/Python editor, guided live coding, projected code traces, debugging cards and LMS submission.
Cross-curricular links	Mathematics: arithmetic operators and practical calculations. English: clear prompts and explanations. Business: simple shopping totals.
External Assessment Integration	Informal coding demonstrations, testing and debugging; official Week 12 Programming assessment details not available.
Wellbeing / ATL Skills	Perseverance, problem-solving, collaboration, self-management, reflection and accessible participation.
Standards	Provisional continuation of prior Python learning; official Grade 12 Programming Week 12 standards not supplied.
Inquiry / TOK Question	How do numeric conversion and testing help a Python program give reliable results?
Success Criteria (Measurable)	MUST: Recognise arithmetic operators and numeric variables. SHOULD: Build a working calculation program using input and conversion. COULD: Debug, test different inputs and improve the program.
National Identity Domain	UAE innovation and responsible digital problem-solving: create fictional everyday calculators for local shopping or travel contexts.